

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "[Guide to Using Scripter Modules](#)" document.

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## Device Specifications

Device Type: Other  
Manufacturer: APC  
Firmware Version: N/A  
Model(s): AP9630, AP9631, AP9635

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scripter Version |
|----------------------------------------|-------------------------|
| 3.01.0000-b010                         | 2.1.0                   |

## Version History

| Module Version | Date      | Notes              |
|----------------|-----------|--------------------|
| 1_0_1_0        | 2/13/2019 | Updated comm sheet |
| 1_0_0_0        | 5/10/2018 | Initial Version    |

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## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly.  
devicePassword and deviceUsername default values are 'apc' and 'apc' respectively.  
Example: `InterfaceName.deviceUsername = 'apc'`  
`InterfaceName.devicePassword = 'apc'`

## Supported Classes and Examples

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                       |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='AP9630')</code>               |
| <b>SerialOverEthernetClass</b>                                                                           |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='AP9630')</code> |
| <b>EthernetClass</b>                                                                                     |
| <code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='AP9630')</code>             |

## Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                                |                                                    |                                                        |                                                             |
|------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------|
| <b>Command</b><br><b>OutletControl</b>                                                         | Value<br>'On'<br><br>'Delay Off'<br><br>'Reboot'   | Value<br>'Off'<br><br>'Delay Reboot'<br><br>'Shutdown' | Value<br>'Delay On'<br><br>'Delay Shutdown'<br><br>'Cancel' |
| <b>Qualifier Key</b><br>'Group'                                                                | Qualifier Value<br>'1' – '3'                       |                                                        |                                                             |
| # OutletControl example<br>InterfaceName.Set('OutletControl', 'On', {'Group': '1'})            |                                                    |                                                        |                                                             |
| <b>Command</b><br><b>UniversalIOControl</b>                                                    | Value<br>'Open'                                    | Value<br>'Close'                                       |                                                             |
| <b>Qualifier Key</b><br>'Port'                                                                 | Qualifier Value<br>'1' – '2'                       |                                                        |                                                             |
| # UniversalIOControl example<br>InterfaceName.Set('UniversalIOControl', 'Open', {'Port': '1'}) |                                                    |                                                        |                                                             |
| <b>Command</b><br><b>UPSControl</b>                                                            | Value<br>'On'<br><br>'Grace Reboot'<br><br>'Sleep' | Value<br>'Off'<br><br>'Grace Sleep'                    | Value<br>'Grace Off'<br><br>'Reboot'                        |
| # UPSControl example<br>InterfaceName.Set('UPSControl', 'On')                                  |                                                    |                                                        |                                                             |

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, BatteryCapacity, Current, InternalTemperature, RuntimeRemaining, and Voltage do not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                          |                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Command</b><br><b>BatteryCapacity</b>                                                                                                                                 | <b>Value</b><br>0 to 100 in steps of 1 %                   |
| # BatteryCapacity examples<br>Value = InterfaceName.ReadStatus('BatteryCapacity')<br>InterfaceName.SubscribeStatus('BatteryCapacity', None, FeedbackHandler)             |                                                            |
| <b>Command</b><br><b>ConnectionStatus</b>                                                                                                                                | <b>Value</b><br>'Connected' <b>Value</b><br>'Disconnected' |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)          |                                                            |
| <b>Command</b><br><b>Current</b>                                                                                                                                         | <b>Value</b><br>Amps                                       |
| # Current examples<br>Value = InterfaceName.ReadStatus('Current')<br>InterfaceName.SubscribeStatus('Current', None, FeedbackHandler)                                     |                                                            |
| <b>Command</b><br><b>DeviceStatus</b> <sup>1</sup>                                                                                                                       | <b>Value</b><br>None                                       |
| # DeviceStatus examples<br>InterfaceName.Update('DeviceStatus')                                                                                                          |                                                            |
| <b>Command</b><br><b>InternalTemperature</b>                                                                                                                             | <b>Value</b><br>'String'                                   |
| # InternalTemperature examples<br>Value = InterfaceName.ReadStatus('InternalTemperature')<br>InterfaceName.SubscribeStatus('InternalTemperature', None, FeedbackHandler) |                                                            |
| <b>Command</b><br><b>RuntimeRemaining</b>                                                                                                                                | <b>Value</b><br>'String'                                   |
| # RuntimeRemaining examples                                                                                                                                              |                                                            |

## Global Scripter Module Communication Sheet

|                                                                                                                                                              |                            |                             |                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------|------------------------------|
| <pre>Value = InterfaceName.ReadStatus('RuntimeRemaining') InterfaceName.SubscribeStatus('RuntimeRemaining', None, FeedbackHandler)</pre>                     |                            |                             |                              |
| <b>Command</b><br><b>Voltage</b>                                                                                                                             | Value<br>VAC               |                             |                              |
| <b>Qualifier Key</b><br><b>'Type'</b>                                                                                                                        | Qualifier Value<br>'Input' | Qualifier Value<br>'Output' | Qualifier Value<br>'Battery' |
| <pre># Voltage examples Value = InterfaceName.ReadStatus('Voltage', {'Type': 'Input'}) InterfaceName.SubscribeStatus('Voltage', None, FeedbackHandler)</pre> |                            |                             |                              |

<sup>1</sup> Calling Update function with DeviceStatus will provide updates for all other commands including BatteryCapacity, Current, InternalTemperature, RuntimeRemaining, and Voltage.

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## Cable and Adapter Requirements

Contact manufacturer for cable and adapter requirements

## Notes for the Device

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### Serial communication

**Port Type:** RS-232

**Baud Rate:** 9600

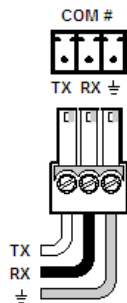
**Data Bits:** 8

**Parity:** None

**Stop Bits:** One

**Flow Control:** None

### Pin Assignments Diagram



Contact manufacturer for pin assignment diagram

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## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                       |              |
|---------------------------------------|--------------|
| <b>Port Type:</b>                     | Ethernet     |
| <b>Default Port:</b>                  | 23           |
| <b>Logon Credentials Supported:</b>   | Yes          |
| <b>Default Username:</b>              | apc          |
| <b>Default Password:</b>              | apc          |
| <b>Multi-Connection Capabilities:</b> | Undetermined |
| <b>Port Changeability:</b>            | Yes          |

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## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device

## Appendix A. Set Commands

|                                       |                               |
|---------------------------------------|-------------------------------|
| Outlet Control Cancel Group 1         | ups -o1 Cancel\x0D\x0A        |
| Outlet Control Delay Off Group 1      | ups -o1 DelayOff\x0D\x0A      |
| Outlet Control Delay On Group 1       | ups -o1 DelayOn\x0D\x0A       |
| Outlet Control Delay Reboot Group 1   | ups -o1 DelayReboot\x0D\x0A   |
| Outlet Control Delay Shutdown Group 1 | ups -o1 DelayShutdown\x0D\x0A |
| Outlet Control Off Group 1            | ups -o1 Off\x0D\x0A           |
| Outlet Control On Group 1             | ups -o1 On\x0D\x0A            |
| Outlet Control Reboot Group 1         | ups -o1 Reboot\x0D\x0A        |
| Outlet Control Shutdown Group 1       | ups -o1 Shutdown\x0D\x0A      |
| Outlet Control Cancel Group 3         | ups -o3 Cancel\x0D\x0A        |
| Outlet Control Delay Off Group 3      | ups -o3 DelayOff\x0D\x0A      |
| Outlet Control Delay On Group 3       | ups -o3 DelayOn\x0D\x0A       |
| Outlet Control Delay Reboot Group 3   | ups -o3 DelayReboot\x0D\x0A   |
| Outlet Control Delay Shutdown Group 3 | ups -o3 DelayShutdown\x0D\x0A |
| Outlet Control Off Group 3            | ups -o3 Off\x0D\x0A           |
| Outlet Control On Group 3             | ups -o3 On\x0D\x0A            |
| Outlet Control Reboot Group 3         | ups -o3 Reboot\x0D\x0A        |
| Outlet Control Shutdown Group 3       | ups -o3 Shutdown\x0D\x0A      |
| UPS Control Grace Off                 | ups -c graceoff\x0D\x0A       |
| UPS Control Grace Reboot              | ups -c gracereboot\x0D\x0A    |
| UPS Control Grace Sleep               | ups -c gracesleep\x0D\x0A     |
| UPS Control Off                       | ups -c off\x0D\x0A            |
| UPS Control On                        | ups -c on\x0D\x0A             |
| UPS Control Reboot                    | ups -c reboot\x0D\x0A         |
| UPS Control Sleep                     | ups -c sleep\x0D\x0A          |
| Universal IO Control Close Port 1     | uio -rc 1 close\x0D\x0A       |
| Universal IO Control Open Port 1      | uio -rc 1 open\x0D\x0A        |
| Universal IO Control Close Port 2     | uio -rc 2 close\x0D\x0A       |
| Universal IO Control Open Port 2      | uio -rc 2 open\x0D\x0A        |

## Appendix B. Update Commands

|               |                        |
|---------------|------------------------|
| Device Status | detstatus -all\x0D\x0A |
|---------------|------------------------|